

can be treated as an advantage as it reduces time to deploy new build on the server. Every small change must be deployed after it is committed or pushed. If the change was made by mistake then an older version is restored in the repository.

Staging Environment:

Features considered fairly stable are merged into staging branch which can be automatically deployed on staging environment as required. Various types of testing is performed to ensure stability of the software. It is very helpful to have separate branch known as staging to have staging environment associated to it.

It helps developers to have multiple branches with different features that can be deployed to the same server simultaneously, simply by syncing/merging code with the local branch that is required on staging environment.

Production Environment:

After successful completion of testing phase, new code base can be deployed on production environment. It should be merged into a stable/properly working branch if the feature(s) was implemented in some other branch. Incomplete features, verbose logging or debugger breakpoints can be identified by taking diff between the production and development branch. Once the diff review is completed, new version of code can be deployed on the production environment. After successful deployment on production server, proper automation process should be implemented to check if all the features of the fix are working properly.

Rollback Strategies:

Sometimes things break at the time of deployment on any server. So rollback is required for maintaining business flow. Issues in rollback will wreak more havoc than the issue due to which rollback is required. If defect is related to code then files can be restored from the backup location. Usually a release introduces some changes or new database structure that is incompatible with the old release. In this case, the application will break after rollback. Proper database backup is restored for proper functioning of application. Backup policy should be implemented to ensure recovery of file to a last known working stage. Normal deployment mechanism is used for rollback also to avoid any cluttered production boxes. Maintaining database and configuration management helps to rollback application smoothly.